SEFI/GWFM: Unified Geometric Field Theory

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1 Worldline Foundations (GWFM)

1.1 Introduction

Quantum mechanics and quantum field theory accurately predict experimental outcomes but leave open the ontological status of particles, fields, and superposition. The Geometric Worldline Field Model (GWFM) begins with a geometric premise: a continuous worldline entity spans spacetime, and the electron field is the projection of this structure. Superposition, interference, and measurement arise from the geometry of this worldline rather than from separate interpretive rules.

1.2 Continuous Worldline Entity

GWFM posits that the electron field arises from a single continuous worldline entity embedded in spacetime. Let M be a spacetime manifold equipped with metric $g_{\mu\nu}$. The entity's worldline is defined as a continuous map

$$y : \mathbb{R} \to M,$$

where the parameter T indexes the full geometric extent of the entity. The worldline spans all allowed paths consistent with the worldline action, producing a dense geometric ensemble whose projection into spacetime yields the appearance of a continuous electron field.

Forward-time segments satisfy $\frac{dt}{dT} > 0$ (electrons), while backward-time segments satisfy $\frac{dt}{dT} < 0$ (positrons). This orientation rule provides a geometric interpretation of particle–antiparticle duality.

Superposition arises from the entity spanning multiple geometric branches, interference arises from overlap regions of the worldline, and measurement corresponds to boundary-condition selection of specific segments.

1.3 Dense Trajectory Ensemble

The worldline spans all allowed trajectories, forming a dense geometric set. Presence-density is defined as

$$p(x) = \int Dy\, D\xi\, e^{S_{\text{entity}}[y,\xi]}\, \delta(x - y(T)).$$

1.4 Field Emergence

The electron field is the projection of the worldline ensemble:

$$\phi(x) = \int dT\, \delta(x - y(T)).$$

1.5 Spin Structure

Spin degrees of freedom are encoded using Grassmann variables $\xi^\mu(T)$ with action

$$S_{\text{spin}} = \int dT\, \xi_\mu \dot{\xi}^\mu.$$

1.6 Gauge Coupling

Electromagnetic coupling is introduced through

$$S_{\text{EM}} = q \int dT\, \dot{y}^\mu A_\mu(y(T)).$$

1.7 Interpretation

Interference corresponds to geometric overlap, measurement corresponds to boundary selection, and identity is global and geometric.

2 Layered Field Identity (SEFI)

2.1 Introduction

SEFI extends GWFM by introducing layered field identity: FIELD ORIGIN, FIELD AUTHORSHIP, FIELD SOVEREIGNTY, WARP EXPRESSION, and the R-LAYER (Reality Field). These layers define stability, identity, and transformation rules.

2.2 FIELD ORIGIN

FIELD ORIGIN defines primitive geometric identity. Stability is treated as a primitive property:

$$\left\| \frac{d^2 r}{dt^2} \right\| \leq S_0,$$

where S_0 is the sovereignty-bound acceleration limit.

2.3 FIELD AUTHORSHIP

Authorship layers contribute geometric identity constructs:

$$F = \sum_{i=1}^{N} A_i(r(t), K(t)),$$

where $K(t)$ is curvature.

2.4 FIELD SOVEREIGNTY

Sovereignty ensures invariance under allowed transformations:

$$F(T(r(t))) = F(r(t)).$$

2.5 WARP EXPRESSION

Warp operators modify worldline geometry while preserving identity:

$$r'(t) = W(r(t)), \qquad F' = F(W(r(t))).$$

2.6 Reality Field Layer (R-Layer)

The R-layer defines the actual geometric manifestation of the entity's worldline into spacetime. It maps worldline geometry, curvature, torsion, stability constraints, sovereignty invariants, and warp transformations into the observable field.

2.7 Mathematical Formulation

Curvature is defined as

$$K(t) = \frac{\|r'(t) \times r''(t)\|}{\|r'(t)\|^3}.$$

The full SEFI identity is

$$F(t) = O(r(t)) + A(r(t), K(t)) + S(r(t)) + W(r(t)) + R(r(t)).$$

3 Photonic QEC from SEFI/GWFM Geometry

3.1 Introduction

Photonic systems suffer from loss, dephasing, mode mixing, and spectral drift. SEFI/GWFM provides a geometric foundation for photonic QEC, defining stability surfaces, geometric stabilizers, and syndrome geometry.

3.2 Stability Surfaces

Sovereignty defines a stability surface: a geometric manifold representing allowed transformations. Errors correspond to displacements orthogonal to this surface.

3.3 Photonic Error Geometry

Loss corresponds to worldline termination, dephasing to torsion, mode mixing to displacement into neighboring surfaces, and spectral wandering to translation along the spectral axis.

3.4 Geometric Stabilizers

Stabilizers preserve sovereignty through curvature, torsion, spectral, and polarization constraints.

3.5 Syndrome Geometry

Weak or QND measurements sample curvature deviation, torsion deviation, spectral displacement, and polarization rotation.

3.6 SPDC Geometry

SPDC annular geometry corresponds to a stability surface defined by FIELD ORIGIN and FIELD SOVEREIGNTY. Diametric correlations arise from sovereign geometric symmetry.

3.7 CHSH Geometry

Polarization evolution is treated as a worldline on the Poincaré-Bloch sphere. Bell states correspond to minimal torsion worldlines.

4 Conclusion

SEFI/GWFM provides a unified geometric field theory. GWFM defines worldline foundations, SEFI defines layered identity, and photonic QEC geometry provides applications. Together they form a coherent geometric interpretation of field behavior, stability, and transformation.